WORKED SOLUTION

STEP I 2001, Question 2

Pure • Algebraic Inequalities

STEP I 2001, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q2

The key idea

With an inequality you can't multiply through by something whose sign you don't know, so the safe move in (i) is to collect everything on one side and factorise: then the answer is read off from a sign diagram. In (ii) the square roots tempt us to square, which is legitimate only when both sides are known to be non-negative. Keeping track of that condition is the whole question.

Solution

Part (i)

We can't multiply by x, since x may be negative. Instead we multiply by x^2, which is strictly positive because $x \neq 0$, so the inequality keeps its direction:

$$\begin{aligned} 1 + 2x - x^2 > \frac{2}{x} &\iff x^2 + 2x^3 - x^4 > 2x \\ &\iff 0 > x^4 - 2x^3 - x^2 + 2x \\ &\iff x\left(x^3 - 2x^2 - x + 2\right) < 0. \end{aligned}$$

The cubic factorises by grouping:

$$x^3 - 2x^2 - x + 2 = x^2(x-2) - (x-2) = (x-2)(x^2-1) = (x-2)(x-1)(x+1).$$

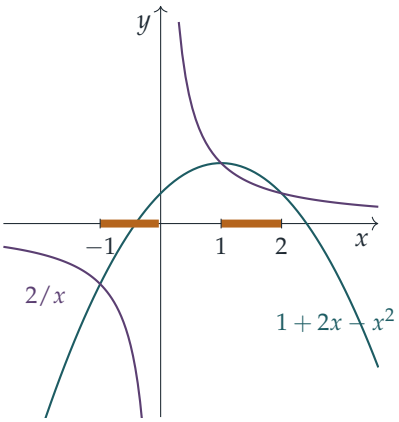
So we need

$$(x+1)\,x\,(x-1)(x-2) < 0.$$

This is a quartic with positive leading coefficient and four simple roots $-1, 0, 1, 2$, so its sign alternates as x crosses each root: positive for $x > 2$, negative on $(1, 2)$, positive on $(0, 1)$, negative on $(-1, 0)$ and positive for $x < -1$. The roots themselves give 0, which is not < 0 (and $x = 0$ is excluded anyway). Hence

$$\boxed{-1 < x < 0 \quad \text{or} \quad 1 < x < 2.}$$

The picture agrees: the parabola $y = 1 + 2x - x^2$ lies above the hyperbola $y = 2/x$ exactly between their crossings at $x = -1$, 1, 2, and on the negative side only until the hyperbola plunges to $-\infty$ at $x = 0$.



Part (ii)

First note that $x \geq -\frac{10}{3}$ gives $x + 4 \geq \frac{2}{3} > 0$ as well as $3x + 10 \geq 0$, so both square roots are defined throughout.

Both sides are non-negative, and for $a, b \geq 0$ we have $a > b \iff a^2 > b^2$. So we may square without gaining or losing solutions:

$$\begin{aligned} \sqrt{3x+10} > 2 + \sqrt{x+4} &\iff 3x + 10 > 4 + 4\sqrt{x+4} + (x+4) \\ &\iff 2x + 2 > 4\sqrt{x+4} \\ &\iff x + 1 > 2\sqrt{x+4}. \end{aligned}$$

Ask yourself Can we simply square again?

Not straight away. The right-hand side $2\sqrt{x+4}$ is positive, but $x + 1$ might be negative, and then squaring could turn a false inequality into a true one. So we split into cases.

- If $x + 1 \leq 0$, the left-hand side is at most 0, while the right-hand side is strictly positive. The inequality fails.
- If $x + 1 > 0$, both sides are positive, so squaring is again reversible:

$$\begin{aligned} x + 1 > 2\sqrt{x+4} &\iff x^2 + 2x + 1 > 4x + 16 \\ &\iff x^2 - 2x - 15 > 0 \\ &\iff (x-5)(x+3) > 0, \end{aligned}$$

  which holds when $x > 5$ or $x < -3$. Combined with $x > -1$, this leaves $x > 5$.

Therefore

$$\boxed{x > 5.}$$

Checks

- At $x = 5$: $\sqrt{25} = 5$ and $2 + \sqrt{9} = 5$, so equality holds there, as it should at the boundary. At $x = 12$: $\sqrt{46} \approx 6.78 > 2 + 4 = 6$. At $x = 0$: $\sqrt{10} \approx 3.16 < 4$.
- In (i), $x = 1.5$ gives $1.75 > 1.33$ and $x = -0.5$ gives $-0.25 > -4$, both true; $x = 0.5$ gives $1.75 < 4$, false.
- A computer tested both inequalities at 2×10^5 random points each, plus points within 10^{-6} of every boundary, and found no disagreement with the answers above.

Common slips

- Multiplying (i) by x without splitting into $x > 0$ and $x < 0$. This loses the interval $(-1, 0)$ or reverses the answer there.
- Squaring $x + 1 > 2\sqrt{x+4}$ without requiring $x + 1 > 0$. That gives the spurious interval $-\frac{10}{3} \leq x < -3$; try $x = -3.2$ in the original to see that it fails ($\sqrt{0.4} \approx 0.63 < 2 + \sqrt{0.8}$).

Beyond STEP

The principle in (ii), that an operation preserves an inequality only if it is an increasing function on the values involved, is the start of how analysts handle inequalities: $t \mapsto t^2$ is increasing on $[0, \infty)$ but not on $\mathbb{R}$. Sign diagrams for products generalise to the fact that a continuous function can change sign only by passing through zero, the intermediate value theorem. Abbott's *Understanding Analysis* treats both carefully.

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